SHEETBASE

Electronic Lab Notebook Product Specification

A flexible, structured and traceable workspace for modern scientific work.

Version 0.1

14 July 2026

Status: Product definition / design baseline

Working product name: Sheetbase ELN

Primary audience: Product, design, engineering, quality, validation and scientific stakeholders

Important: This specification defines product capabilities and design controls. It does not itself constitute a validated state or a claim of regulatory compliance. Validation, procedures, infrastructure qualification and intended-use assessment remain deployment-specific responsibilities.

Document control

Field	Value
Document owner	Product lead
Contributors	Product design, engineering, scientific users, quality assurance
Version	0.1
Date	14 July 2026
Status	Initial product specification
Review cycle	Update at each major product scope or architecture decision
Intended use of this document	Product definition, delivery planning, design review and validation planning input

Decision summary

Decision	Baseline
Product model	Document-first ELN with structured records and data blocks underneath.
Hierarchy	Workspace > Project > Folder > item. Folders are nested.
Experiments	Simple structured records with a primary free-form notebook page.
Editor	Block-based, free-form and collaborative; not a fixed laboratory form.
Tables	Both local inline tables and shared database-backed tables.
Graphing	Dedicated analysis studio and embeddable, versioned chart blocks.
Compliance	Designed to support controlled GxP use, including audit trails, review, signatures and locking.
Navigation	Attio-inspired project tree and saved views; fully configurable navigation is a later capability.
Target market	Biotechnology first, but the core model remains discipline-neutral.

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1. Executive summary

Sheetbase ELN is a web-first electronic lab notebook that combines a flexible scientific document editor with structured records, shared data tables, full-featured graphing and controlled review workflows. It is intended to replace fragmented combinations of paper notebooks, spreadsheets, shared drives and disconnected analysis tools without forcing scientists into rigid forms.

The product should feel calm, precise and information-dense. The primary interaction is a notebook page composed from movable blocks: text, sections, tables, charts, files, protocols, calculations and linked scientific records. Projects and nested folders organise work in an Attio-inspired sidebar, while saved views provide cross-project operational perspectives such as My experiments, Awaiting review and Failed runs.

The system must work well for exploratory research and controlled GxP workflows. Compliance features are therefore built into the record lifecycle rather than added as a separate document-management layer. Version history, audit events, review, electronic signatures, data lineage, access controls and record locking are first-class capabilities.

Product proposition

A modern ELN that gives scientists the freedom of a document, the structure of a database and the traceability required for reliable science.

1.1 Product outcome

- Scientists can capture experimental context, methods, observations, data and conclusions in one coherent record.
- Structured data remains reusable across experiments, projects, dashboards and analysis views.
- Charts and calculations are reproducible because the source, transformation, settings and engine version are retained.
- Quality reviewers can understand what changed, review supporting evidence and sign a precise record version.
- Organisations can configure the system for research or controlled use without maintaining separate products.

1.2 Positioning

Sheetbase ELN sits between general-purpose collaborative documents and heavyweight laboratory informatics platforms. It should be easier to adopt than a LIMS, more scientifically capable than a generic workspace, and more traceable than standalone spreadsheets or analysis files. It is not intended to replace every specialist instrument system or enterprise LIMS; it provides the connective workspace in which scientific work is planned, recorded, analysed, reviewed and linked.

2. Product vision and principles

Vision: make structured, reproducible scientific work feel as natural as writing a note.

Principle	Implication
Document first	The page is the primary working surface. Structure should support the scientist rather than interrupt them.
Structured underneath	Experiments, samples, protocols, tables and datasets are linkable records with stable identifiers.
Traceability by default	Sources, changes, calculations, file hashes, versions and authorship remain reconstructable.
Progressive control	Exploratory work is lightweight; controlled work adds required fields, review, signatures and locking.
Calm density	Fit substantial scientific content on screen without oversized typography or excessive empty space.
Reproducible analysis	A chart is a saved analysis specification, not merely a picture.
Open and portable	Data is accessible through export, API and documented formats; customers are not trapped.
Configuration over custom code	Statuses, fields, templates, views and workflows should be configurable where safe.

3. Scope, goals and non-goals

3.1 Goals

- Provide a polished, desktop-first ELN experience for biotechnology and adjacent scientific teams.
- Support free-form pages with text, sections, multiple tables, graphs, images, files and linked records.
- Provide nested project folders and cross-project saved views.
- Treat experiments as simple records, avoiding an over-engineered mandatory schema.
- Support both inline page tables and reusable shared database tables.
- Provide graphing capable of routine scientific and publication-quality work.
- Support controlled review, version comparison, electronic signatures, record locking and audit inspection.
- Expose a stable API and integration model for instruments, LIMS, data platforms and automation.
- Remain understandable and useful outside biotechnology by keeping core entities generic and customisable.

3.2 Non-goals for the first production release

- A complete enterprise LIMS covering scheduling, batch disposition, inventory procurement and manufacturing execution.
- A replacement for every specialist statistical package, image-analysis platform or instrument control application.
- Unrestricted no-code object modelling in the first release; core record types are curated initially.
- Pixel-positioned canvas design. Pages use an ordered block layout for accessibility, collaboration and reliable version comparison.
- Full offline operation. Resilient local drafts are allowed, but controlled records require a trusted server connection.
- Regulatory certification. The product supplies capabilities and evidence to support a customer validation programme.

3.3 Deployment assumptions

- Primary delivery is multi-tenant SaaS, with a private-cloud or single-tenant option for regulated customers.
- The application is browser-based and optimised for desktop screens; tablet viewing and light editing are supported.
- PostgreSQL is the transactional system of record and object storage holds raw files, exports and snapshots.
- Identity is normally delegated to an organisation identity provider through OIDC or SAML.
- English is the initial interface language, but the data model and UI should be localisation-ready.

4. Users, roles and core use cases

Persona	Primary needs
Scientist	Plans and records work, enters observations and data, links samples and files, creates charts, submits records for review.
Project lead	Organises projects, monitors progress, standardises templates, reviews linked work and manages milestones.
Data analyst	Builds reusable datasets, calculations and charts; creates saved views and exports analysis-ready data.
Reviewer / QA	Reviews exact versions, compares changes, comments, verifies checks, signs and locks controlled records.
Laboratory manager	Monitors workload, sample status, overdue reviews, data quality and project dashboards.
Workspace administrator	Manages users, teams, permissions, fields, statuses, retention, integrations and compliance settings.
Integration engineer	Uses APIs, service accounts and webhooks to connect instruments and enterprise systems.

4.1 Core end-to-end use cases

1. Create a project, organise folders and start an experiment from a blank page or approved template.
2. Record objectives, materials, steps, observations, images and instrument files in a free-form page.
3. Enter data in an inline table, then promote it to a shared table when it becomes reusable.
4. Build a chart from a page table, shared table, dataset or imported file and embed the saved analysis in the notebook.
5. Link samples, protocols and datasets to the experiment without duplicating their master information.
6. Collaborate using comments, mentions, assignments and real-time presence while preserving authorship.
7. Submit a version for review, compare changes, resolve comments, apply electronic signatures and lock the approved record.
8. Search across content, properties and table values, then save a filtered operational view.

9. Export a complete experiment package containing the signed record, audit trail, attachments and structured data.
10. Connect an instrument or data platform through APIs and retain source lineage for imported and derived data.

5. Information architecture and navigation

The information architecture is deliberately simple: the workspace contains projects; projects contain nested folders and items. Items may also be accessed through saved views, links and search without changing their canonical location.

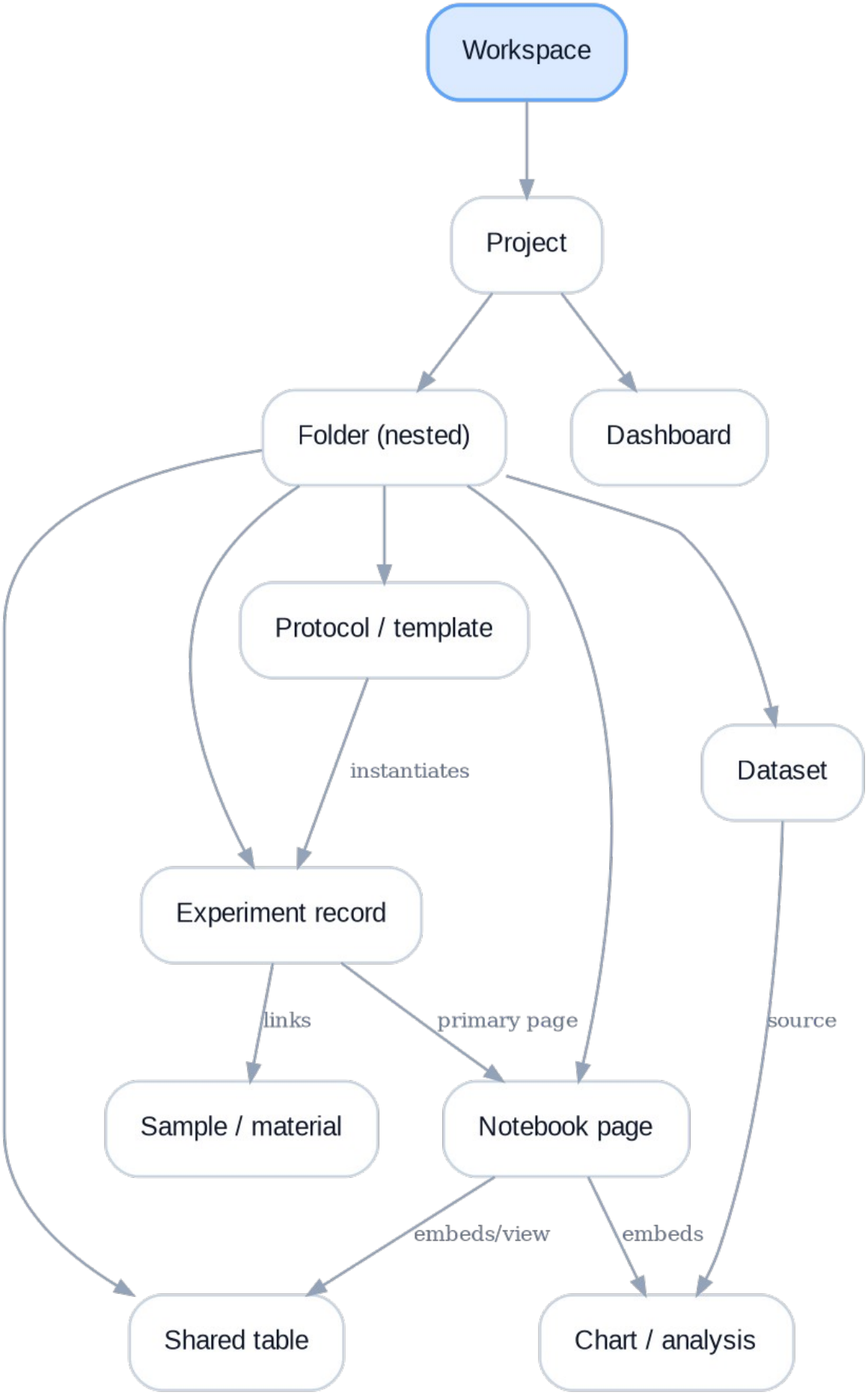

Workspace
Project
Folder (nested)
Dashboard
Protocol / template
instantiates
Dataset
Experiment record
links
primary page
source
Sample / material
Notebook page
embeds/view
embeds
Shared table
Chart / analysis

Figure 1. Proposed information architecture.

5.1 Global application shell

- Left sidebar: workspace switcher, Home, My work, Activity, Tasks, Reviews, project tree and saved views.
- Top bar: breadcrumbs, universal search, create action, share controls, help and user settings.
- Central canvas: the selected page, table, dashboard, registry or analysis workspace.
- Right inspector: context-sensitive properties, activity, comments, linked records and compliance details.
- Command palette: keyboard-first navigation and creation using a single global shortcut.

5.2 Project sidebar

- Projects are collapsible and may be pinned, reordered or archived.
- Folders may be nested to at least five levels and contain mixed item types.
- Common project areas such as Overview, Experiments, Protocols, Samples, Datasets, Analysis and Reports are provided as optional starter folders, not immutable system sections.
- Users may drag items between folders subject to permission and lifecycle controls.
- Counters may show review tasks, failed checks or overdue items without making the tree visually noisy.
- A later configurable-navigation release allows administrators to define default project structures and object-specific icons.

5.3 Saved views

- Personal and shared views can filter across projects by record type, owner, status, tag, date, linked record and custom field.
- Initial system views: My experiments, Recently edited, Awaiting review, Locked entries, Failed checks, Templates and Expiring materials.
- Views support table, kanban, calendar, timeline and gallery presentations where the underlying fields permit them.
- A view is a saved query and presentation configuration, not a copy of records.

6. Core domain model

The domain model separates working documents from reusable structured records. This avoids forcing every observation into a database row while still making critical data discoverable and reusable.

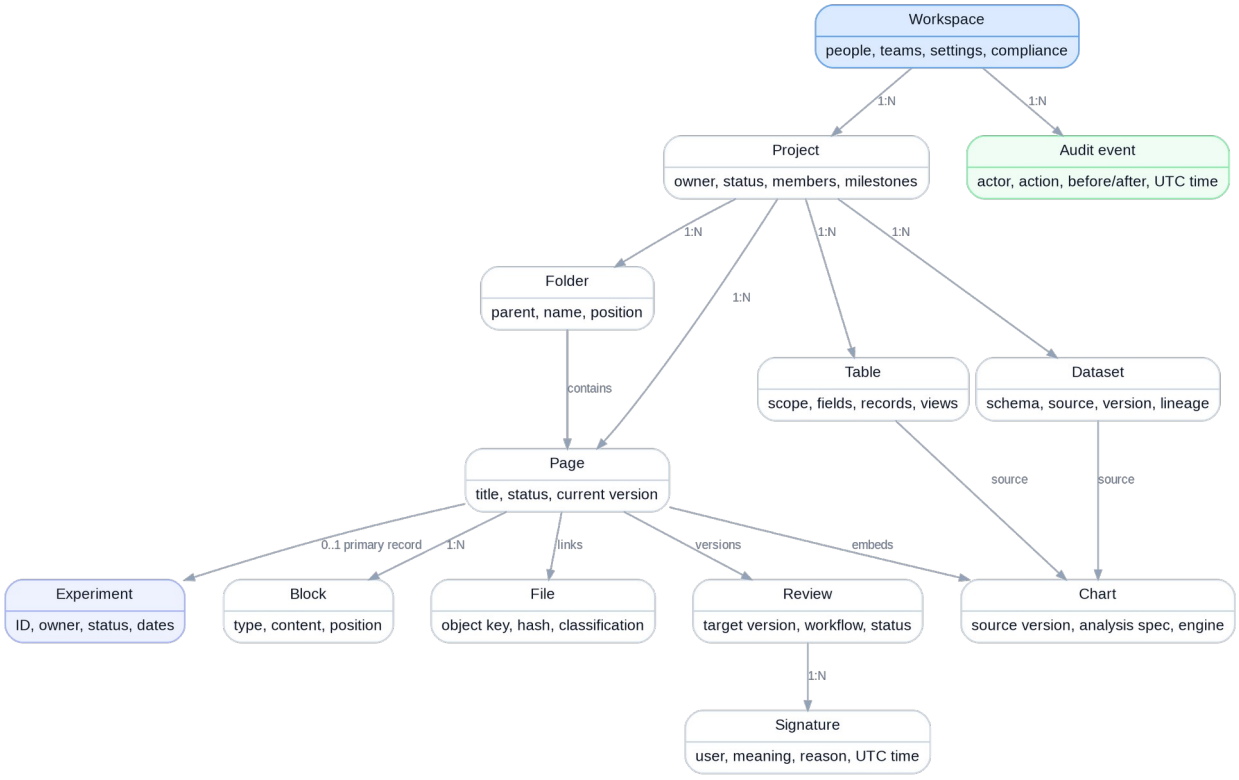


Figure 2. Simplified core data model.

Entity	Definition
Workspace	Tenant boundary containing people, teams, configuration, security and compliance settings.
Project	Top-level scientific initiative with owner, status, members, milestones and dashboard.
Folder	Nested organisational node containing mixed item types.
Page	Versioned block document. May stand alone or serve as the primary page for an experiment.
Block	Ordered unit of content such as text, table, chart, image, equation or linked record.
Experiment	Simple record containing identity, status, ownership, dates, tags, project and a primary page link.
Protocol	Versioned method or reusable template composed from steps, materials and content blocks.
Sample / material	Customisable inventory or reference record with lineage, location and linked usage.
Table	Local or shared structured dataset with typed fields, records, validation and views.
Dataset	Versioned imported or derived data asset with provenance and schema.
Chart	Saved analysis specification linked to a source version and embeddable in pages or dashboards.
File	Immutable binary object with hash, metadata, classification and optional replacement relationship.
Review	Workflow instance bound to a precise page or protocol version.
Signature	Identity, meaning, reason and timestamp linked to a reviewed version.
Audit event	Append-only event recording who performed what action, when and against which entity.

6.1 Experiment record

Experiments intentionally use a small required schema. Organisations may add custom properties, but the free-form page remains the primary scientific record.

Field	Requirement
Experiment ID	Stable, human-readable identifier generated from a configurable pattern.
Title	Required. Editable until controlled locking rules apply.
Status	Required. Uses the configured experiment lifecycle.
Owner	Required individual. Contributors are many-to-many.
Project and folder	Required project; folder may be optional.
Start and end dates	Optional in exploratory work; configurable as required.
Tags and custom fields	Optional, workspace-configured.
Primary page	Exactly one canonical notebook page.
Linked records	Samples, protocols, datasets, charts, files and related experiments.

7. Product experience specification

7.1 Home and My work

- Home provides recent projects, recent pages, assigned work, review requests and quick creation.
- My work is a personal operational view covering owned experiments, assigned tasks, drafts, mentions and due reviews.
- Cards are compact and actionable; the screen should prioritise work, not decorative analytics.
- Users may customise visible sections without changing the underlying records.

7.2 Project overview

- Header contains project title, status, phase, description, owner, members, tags and share controls.

- Summary cards show counts and exceptions: active experiments, linked samples, datasets, protocols, pending reviews and failed checks.
- Optional milestone timeline provides project-level dates and stages.
- Recent items, key charts and activity feed are presented as configurable widgets.
- The right inspector exposes project properties, pinned links, compliance state and quick actions.

7.3 Notebook and experiment editor

The editor is a linear block document with optional sections and two-column layouts. It should feel immediately writable while retaining stable block identities for comments, permissions, version comparison and auditability.

- A page header provides title, record type, status, identifier, owner, date, contributors, templates and lifecycle actions.
- Slash commands and a block inserter create content without exposing complex menus.
- Blocks can be reordered, duplicated, converted, collapsed, nested under sections and moved between compatible pages.
- Users can select multiple blocks for grouped actions and comments.
- Autosave is continuous. The interface clearly distinguishes saved, syncing, offline draft and conflict states.
- The right inspector switches between properties, activity, comments, linked records, files and checks.
- The editor supports keyboard navigation and does not rely on hover-only controls.

7.4 Block library

Block category	Capabilities
Text	Paragraph, heading, list, checklist, quote, callout and divider.
Scientific notation	Equation, inline symbols, superscript/subscript, units and code.
Section	Named, collapsible container with optional description and completion state.
Table	Inline local table or view of a shared table/dataset.
Chart	Saved chart or new analysis created from a selected source.
File and media	File attachment, image, PDF preview, gallery and external embed.
Linked record	Experiment, sample, protocol, dataset, chart, task or person reference.
Protocol step	Numbered instruction with materials, parameters, timer and completion.
Calculation	Formula or analysis result with inputs, units and stored result metadata.
Decision / result	Structured callout for conclusion, deviation, issue or outcome.
Signature / review summary	Read-only controlled block showing signature and review state.
Dashboard widget	Compact metric, chart, filtered list or status summary.

7.5 Table system

Tables are a core product surface rather than simple document grids. The user may choose a page-owned inline table or a reusable shared table. An inline table can be promoted to a shared table while preserving references and history.

Capability	Specification
Field types	Text, rich text, number, unit-aware number, date/time, checkbox, select, multi-select, person, status, formula, relation, rollup, file, barcode and identifier.
Validation	Required, allowed values, range, regex, uniqueness, unit compatibility and custom rule.
Views	Table, kanban, calendar, timeline, gallery and compact list.
Data operations	Filter, sort, group, pin, freeze, hide, resize, bulk edit, fill, copy/paste, undo and keyboard navigation.
Formulas	Row formulas, aggregate formulas and named calculated fields with clear error states.
Import	CSV and XLSX mapping with preview, type inference, validation summary and repeatable import recipe.
Export	CSV, XLSX and API output with selected fields, view filters and optional metadata.

Capability	Specification
Lineage	Rows and cells retain source/import information when created from controlled datasets.
Scale	Interactive client view for typical tables; server-side query, paging and aggregation for large datasets.

7.6 Graphing and Analysis Studio

Graphing must be substantially more capable than a basic dashboard chart. Every chart is backed by a saved, inspectable analysis specification. Users can build charts in a dedicated studio and embed the same live or snapshotted chart in notebooks, dashboards and reports.

Area	Specification
Sources	Inline table, shared table view, dataset version, imported file, API query or previous analysis output.
Chart types	Line, scatter, bar, grouped/stacked bar, area, histogram, box, violin, heat map, waterfall, dot plot, plate map and combination charts.
Encodings	X, Y, colour, shape, size, label, facet, series and tooltip.
Transforms	Filter, sort, group, aggregate, pivot, unpivot, derive, normalise, baseline, fold change, log transform and missing-value handling.
Statistics	Count, mean, median, geometric mean, SD, SEM, confidence interval, quantiles, regression, trend line and configurable error bars.
Scientific fits	Linear, polynomial, exponential and logistic curve fits, including 4PL/5PL assay curves where enabled.
Annotations	Reference lines, acceptance bands, labels, notes, thresholds, excluded-point markings and event markers.
Styling	Theme, palette, marker, line style, axes, grid, legend, labels, significant figures and scientific notation.
Reproducibility	Persist source version, transformation graph, formulas, parameters, exclusions, chart specification and analysis engine version.
Export	PNG, SVG and PDF with publication presets, transparent background and 300/600 dpi raster options.
Performance	Downsample or aggregate large data interactively while preserving an exact full-resolution export path.

Chart integrity rule

Changing source data, exclusions, formulas or statistical settings must create a visible chart revision. A pasted image alone is not considered a reproducible chart.

7.7 Protocols and templates

- Protocols are versioned documents with objective, materials, numbered steps, parameters, warnings, attachments and linked records.
- Templates may define a complete page structure, required blocks, default properties, required links and review workflow.
- Using a controlled template records the template version and creates a snapshot so later template updates do not silently alter the experiment.
- Template authors can compare versions, request approval, publish, retire and supersede templates.
- Completed protocol steps may capture performer, timestamp, values and exceptions when used in an experiment.

7.8 Sample and material registry

- A registry provides reusable sample, reagent, construct, batch and material records without making inventory management mandatory.
- Fields include identifier, type, parent/source, batch, concentration and unit, volume, storage location, expiry, owner, status and custom properties.
- Chain-of-custody events record creation, movement, aliquoting, consumption and disposal where enabled.
- Barcode labels and scanning are supported through browser-compatible scanners and mobile camera workflows in a later release.
- The registry may be replaced or synchronised with an external LIMS through the integration layer.

7.9 Files, datasets and provenance

- Files are immutable objects identified by hash. Replacing a file creates a new object and explicit replacement relationship.
- The system stores uploader, timestamp, source, size, MIME type, classification and linked record metadata.
- Raw, intermediate and derived classifications are available; customers may configure retention and permitted transformations.
- Dataset records contain schema, source files, import recipe, transformation history, quality checks and version.
- Preview generation, virus scanning and metadata extraction are asynchronous and never alter the original file.
- Bulk export packages include a manifest of file hashes and relationships.

7.10 Collaboration, activity and tasks

- Real-time presence shows collaborators without obscuring the scientific content.
- Comments may attach to a page, block, table row, cell, chart element or version comparison.
- Mentions create notifications; comments can be assigned, resolved and reopened.
- Tasks may be created from comments or records and carry owner, due date, priority and link back to context.
- Activity feeds summarise meaningful events and can be filtered; raw audit data remains separate and complete.
- Notifications support in-app and email delivery, with Slack or Microsoft Teams integration later.

7.11 Review and version comparison

- Submission creates a stable version and prevents untracked edits to the review target.
- Reviewers compare versions side by side or inline, including text, properties, table cells, linked files and chart specifications.
- Comments and checklist items are linked to the reviewed version and remain visible after resolution.
- Configured review steps may require author, peer reviewer, QA reviewer, approver or witness roles.
- Signing records identity, meaning, reason, UTC timestamp and the exact content hash.
- Approved records may be locked. Amendments create a new controlled version rather than altering the signed version.

7.12 Search and discovery

- Universal search covers titles, page text, properties, table values, comments where authorised and extracted file text where enabled.
- Results are permission-filtered and grouped by item type with highlighted matches.
- Filters include workspace, project, folder, owner, status, date, tag, record type and custom field.
- Users can search identifiers and barcodes directly.
- The command palette combines search, navigation and creation actions.

7.13 Administration and configuration

- Workspace profile, teams, users, domains and authentication settings.
- Roles, permission groups, project defaults and external collaborator policy.
- Custom fields, tags, statuses, identifiers, units and naming conventions.
- Template library, review workflows, signature meanings and controlled record settings.
- Retention, archival, legal hold, backup export and data residency settings where supported.
- API keys, service accounts, webhooks, integrations and rate-limit visibility.
- Audit explorer, security events, release information and validation evidence downloads.

8. Functional requirements

Priority uses Must, Should and Could. Release uses R1 for the first production baseline, R2 for planned expansion and Future for capabilities intentionally deferred. Requirements are product-level and should be decomposed into testable engineering stories.

8.1 Foundation and navigation

ID	Requirement	Priority	Release
NAV-001	The system shall support multiple workspaces with strict tenant separation.	Must	R1
NAV-002	A workspace shall contain projects, people, teams, views and configuration.	Must	R1
NAV-003	Projects shall support nested folders containing mixed item types.	Must	R1
NAV-004	Users shall be able to pin, reorder, rename, move and archive projects and folders subject to permission.	Must	R1

ID	Requirement	Priority	Release
NAV-005	The application shall provide breadcrumbs for all project-contained items.	Must	R1
NAV-006	The application shall provide universal search and a keyboard command palette.	Must	R1
NAV-007	Users shall be able to create personal and shared saved views across projects.	Should	R1
NAV-008	Administrators shall be able to configure default project structures.	Should	R2
NAV-009	Administrators shall be able to configure sidebar sections and record-type navigation.	Could	Future

8.2 Pages and editor

ID	Requirement	Priority	Release
EDT-001	Users shall be able to create a blank page or create one from a template.	Must	R1
EDT-002	Pages shall consist of ordered, independently identifiable blocks.	Must	R1
EDT-003	The editor shall support text, headings, lists, checklists, callouts, equations, code, files, images, tables, charts and linked records.	Must	R1
EDT-004	Users shall be able to add, delete, reorder, duplicate, convert and collapse compatible blocks.	Must	R1
EDT-005	The editor shall support named sections and a two-column layout.	Should	R1
EDT-006	The editor shall autosave and visibly report save and synchronisation state.	Must	R1
EDT-007	Users shall be able to undo and redo changes within a working session.	Must	R1
EDT-008	The system shall preserve block identity across edits and moves.	Must	R1
EDT-009	The editor shall support real-time collaborative editing and presence.	Should	R1
EDT-010	The editor shall provide accessible keyboard navigation and non-hover alternatives.	Must	R1
EDT-011	Pages shall support links to experiments, samples, protocols, datasets, tables, charts, files and people.	Must	R1
EDT-012	A user shall be able to export a page to PDF and a portable structured format.	Must	R1
EDT-013	Users shall be able to create reusable blocks or page fragments.	Could	R2

8.3 Tables and data

ID	Requirement	Priority	Release
TBL-001	The system shall support page-owned inline tables and workspace-shared tables.	Must	R1
TBL-002	An inline table shall be promotable to a shared table without losing data or links.	Must	R1
TBL-003	Tables shall support typed fields, required fields and validation rules.	Must	R1
TBL-004	Tables shall support formula, relation and rollup fields.	Should	R1
TBL-005	Users shall be able to filter, sort, group, hide, freeze and resize fields.	Must	R1
TBL-006	Tables shall support bulk edit, copy/paste and keyboard-first cell navigation.	Must	R1
TBL-007	Shared tables shall support multiple saved views.	Must	R1
TBL-008	The system shall import CSV and XLSX using a mapping and validation preview.	Must	R1
TBL-009	The system shall export view-filtered data to CSV and XLSX.	Must	R1
TBL-010	Tables shall provide row and cell comments with permissions inherited from the table.	Should	R1
TBL-011	Data changes shall be versioned and represented in the audit trail when audit is enabled.	Must	R1
TBL-012	Large tables shall use server-side paging, filtering and aggregation.	Must	R1
TBL-013	Tables shall support configurable unit-aware numeric fields.	Should	R2

8.4 Charting and analysis

ID	Requirement	Priority	Release
CHT-001	Users shall be able to create a chart from an inline table, shared table, dataset or imported file.	Must	R1
CHT-002	Charts shall persist source identity and source version.	Must	R1

ID	Requirement	Priority	Release
CHT-003	Charts shall support line, scatter, bar, histogram, box, heat map and plate map in R1.	Must	R1
CHT-004	Charts shall support facets, multiple series and secondary grouping.	Must	R1
CHT-005	The analysis pipeline shall support filter, derive, aggregate, normalise and log transform operations.	Must	R1
CHT-006	Charts shall support SD, SEM and confidence interval error representations.	Must	R1
CHT-007	Users shall be able to define exclusions with a recorded reason.	Must	R1
CHT-008	Charts shall support configurable axes, labels, legends, reference lines and annotations.	Must	R1
CHT-009	A chart revision shall be created when data, transformation, exclusions or material settings change.	Must	R1
CHT-010	Charts shall be embeddable in pages and dashboards as live or snapshotted views.	Must	R1
CHT-011	Users shall be able to export PNG, SVG and PDF.	Must	R1
CHT-012	The system shall retain analysis engine and formula versions.	Must	R1
CHT-013	The system should support regression and 4PL/5PL curve fitting.	Should	R2
CHT-014	Users should be able to save reusable analysis templates.	Should	R2
CHT-015	The system should provide a documented plug-in interface for advanced analyses.	Could	Future

8.5 Experiments, protocols and records

ID	Requirement	Priority	Release
REC-001	An experiment shall have a stable ID, title, status, owner, project and primary page.	Must	R1
REC-002	The experiment schema shall support configurable custom properties.	Must	R1
REC-003	Experiments shall link to samples, protocols, datasets, charts, files and related experiments.	Must	R1
REC-004	Protocols shall be versioned and may be published as controlled templates.	Must	R1
REC-005	Creating an experiment from a template shall record the source template version.	Must	R1
REC-006	Templates shall define required blocks, fields, links and workflow where configured.	Should	R1
REC-007	Sample records shall support custom types, identifiers, status, storage and lineage links.	Should	R1
REC-008	Sample movement, aliquot and consumption events shall be available as an optional controlled feature.	Should	R2
REC-009	Users shall be able to create custom record types using an administrator-defined schema.	Could	Future

8.6 Files and provenance

ID	Requirement	Priority	Release
FIL-001	Uploaded files shall be stored without modifying the original bytes.	Must	R1
FIL-002	The system shall calculate and retain a cryptographic hash for each stored file.	Must	R1
FIL-003	Replacing a file shall create a new file object and preserve the relationship to the prior file.	Must	R1
FIL-004	Files shall expose uploader, timestamp, size, type, classification and linked records.	Must	R1
FIL-005	Preview generation and metadata extraction shall not alter the original.	Must	R1
FIL-006	Datasets shall retain source, schema, import settings and transformation lineage.	Must	R1
FIL-007	Users shall be able to export a package containing files, record content, structured data, audit trail and manifest.	Must	R1
FIL-008	The platform shall support direct object-storage ingestion through a controlled integration.	Could	R2

8.7 Collaboration, review and compliance

ID	Requirement	Priority	Release
CTL-001	The system shall maintain version history for controlled pages, protocols, tables and charts.	Must	R1
CTL-002	Users shall be able to compare two versions including content, properties, table values and chart settings.	Must	R1

ID	Requirement	Priority	Release
CTL-003	Comments shall be attachable to pages, blocks, records, rows, cells and version differences.	Must	R1
CTL-004	A review request shall bind reviewers to a precise immutable version.	Must	R1
CTL-005	Review workflows shall support configurable stages and role requirements.	Must	R1
CTL-006	Electronic signatures shall include signer, meaning, reason, UTC timestamp and signed content identifier.	Must	R1
CTL-007	A signed version shall not be silently modified or deleted.	Must	R1
CTL-008	Approved records shall be lockable and amendments shall create a new controlled version.	Must	R1
CTL-009	The system shall provide an append-only audit trail of security-relevant and regulated actions.	Must	R1
CTL-010	Audit entries shall include actor, action, entity, timestamp and before/after context where applicable.	Must	R1
CTL-011	Audit events shall use a trusted time source and be displayed in user-local and UTC formats.	Must	R1
CTL-012	The system shall support required reasons for change for configured objects and states.	Must	R1
CTL-013	Authorised users shall be able to export audit trails in human-readable and machine-readable formats.	Must	R1
CTL-014	The system shall support configurable retention, archival and legal-hold policies.	Should	R2
CTL-015	The system shall provide risk-based audit-trail review views and reports.	Should	R2

8.8 Search, automation and administration

ID	Requirement	Priority	Release
ADM-001	Search results shall be filtered by the requesting user's permissions.	Must	R1
ADM-002	Search shall cover titles, text, identifiers, properties and table values.	Must	R1
ADM-003	Users shall receive in-app notifications for mentions, assignments, review requests and workflow changes.	Must	R1
ADM-004	The system shall support configurable email notifications.	Should	R1
ADM-005	Administrators shall manage users, teams, roles, projects, fields, statuses and templates.	Must	R1
ADM-006	Administrators shall configure identifier patterns, units, tags and naming conventions.	Should	R1
ADM-007	The system shall provide service accounts, scoped API tokens and webhooks.	Must	R1
ADM-008	Administrators shall be able to disable, rotate and audit API credentials.	Must	R1
ADM-009	The platform should support workflow automation triggered by record events and conditions.	Should	R2
ADM-010	The platform may provide an AI assistant for summarisation and drafting only when provenance, permissions and customer controls are explicit.	Could	Future

9. Controlled workflows and compliance

Compliance posture

Sheetbase ELN should be designed to support validated use in closed-system environments. Product features alone do not make a customer process compliant; intended use, validation, procedures, training, infrastructure and operational controls must be considered together.

9.1 Regulatory design baseline

The product baseline is informed by 21 CFR Part 11 controls for trustworthy electronic records and signatures, the current EU GMP Annex 11 baseline for computerised systems, and MHRA GxP data-integrity guidance. The European Commission opened a consultation on a revised Annex 11 in 2025; the product should monitor and prepare for the emerging lifecycle, supplier, security and data-integrity expectations without treating a consultation draft as the current legal baseline. [R1-R5]

Part 11 applies to electronic records maintained under FDA record requirements and includes controls for closed systems and electronic signatures. Annex 11 expects the application to be validated, IT infrastructure to be qualified

and risk management to apply through the system lifecycle. MHRA guidance places emphasis on complete, consistent and reconstructable data and activities. [R1-R3, R5]

9.2 GxP mode

GxP mode is a workspace or project policy profile, not a separate application. Controls may be made stricter but not weaker at lower levels.

Control	Exploratory default	GxP mode
Versioning	Continuous history; named versions optional.	Controlled versions required at submission, signature and unlock.
Reason for change	Optional.	Required for configured fields, post-submission edits and administrative overrides.
Review	Optional or lightweight.	Configured workflow with role separation and checklist.
Electronic signature	Available.	Required at defined lifecycle transitions.
Record locking	Optional.	Automatic or required after approval.
Deletion	Soft delete and recovery.	No destructive deletion of controlled records; archive with audit event.
Files	Immutable storage recommended.	Immutable original, hash verification and controlled replacement.
Audit trail	Core security events retained.	Expanded regulated actions, review tools and export.
Permissions	Role and project permissions.	Role separation, restricted overrides and periodic access review support.

9.3 Controlled lifecycle

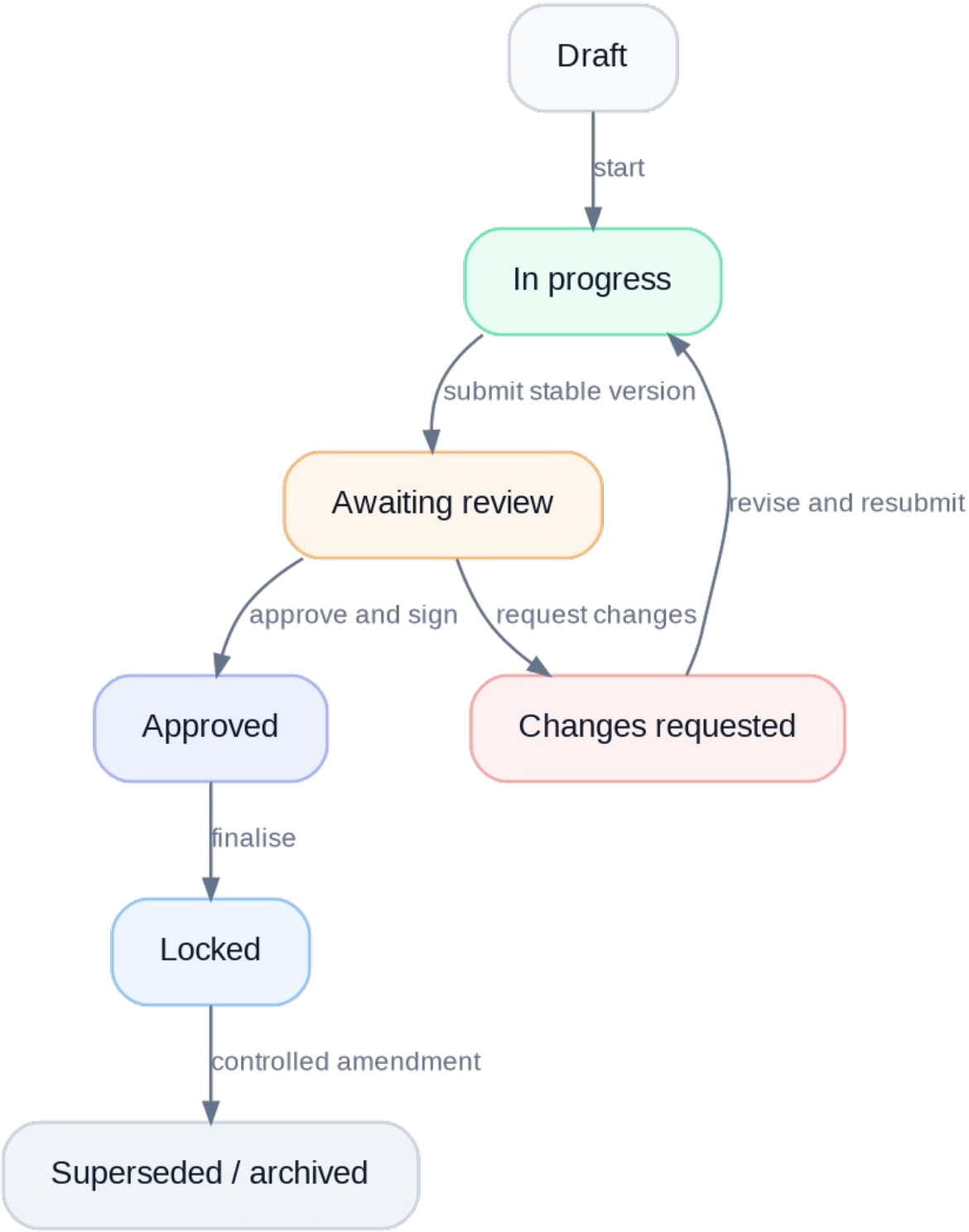


Figure 3. Default controlled experiment lifecycle.

- Draft: record exists but work has not formally started.
- In progress: active scientific work. Changes are attributed and autosaved.
- Awaiting review: a stable version is submitted; subsequent author edits create a new draft rather than changing the review target.
- Changes requested: reviewer response returns work to the author with comments and checklist state.
- Approved: required signatures have been applied to the reviewed version.
- Locked: approved content cannot be edited. Authorised amendment creates a new version and preserves the signed record.
- Superseded or archived: record remains retrievable according to retention and access policy.

9.4 Audit trail

Audit event content	Minimum data
Identity	Event ID, tenant, actor, authentication context and impersonation state if applicable.
Time	Trusted UTC timestamp plus rendered local time.
Action	Create, edit, delete/archive, view where required, export, sign, approve, permission change, configuration change or integration action.
Object	Entity type, stable entity ID, version and project/workspace context.
Change	Before and after value or hashes, changed fields, reason and source application/API.
Integrity	Append-only storage, tamper-evident chaining or equivalent integrity control, retention and verified export.

9.5 Electronic signatures

- Every signature is unique to one user identity and cannot be reassigned.
- The signing action requires re-authentication or an equivalent policy-controlled confirmation.
- The signature captures printed name, user ID, meaning, reason, timestamp and record/version hash.
- The signature display is permanently linked to the record and included in exports.
- Administrators configure allowed signature meanings such as Authored, Reviewed, Approved and Witnessed.
- Signature withdrawal is not destructive; it creates a new audit event and workflow state according to policy.

9.6 Validation support package

- Product requirements and release-level traceability to verification evidence.
- Supplier and architecture information, including component inventory and security documentation.
- Release notes, known issues, change classifications and migration notes.
- Installation/deployment qualification guidance for supported deployment models.
- Configurable export of workspace settings, workflows, roles and controlled templates.
- API documentation and test environment support.
- Customer validation is risk-based and remains the customer's responsibility.

10. Security and permissions

10.1 Role model

Role	Typical authority
Workspace owner	Commercial/tenant authority; manages ownership and critical workspace controls.
Administrator	Users, teams, configuration, integrations and non-content operational administration.
Quality administrator	Controlled templates, workflows, signature policy, lock/unlock and audit review.
Project lead	Project membership, folder organisation, project properties and project-level templates.
Scientist	Create and edit permitted records, data, files and analyses; submit for review.
Reviewer	Read reviewed scope, comment, complete checklist and sign according to assigned role.
Viewer	Read-only access to authorised content and exports.
Service account	API-only principal with explicitly scoped permissions and owner.

10.2 Permission model

- Permissions combine workspace role, team membership, project membership, item ACL and lifecycle state.
- Default inheritance flows from workspace to project to folder to item; authorised users may create explicit exceptions.
- Capabilities include view, comment, edit content, edit data, manage files, submit, review, sign, approve, lock, export, share and administer.
- External collaborators are visibly identified and may be restricted from exports, audit data and cross-project search.
- Permission changes are audited and must not retroactively hide a signature or audit event from authorised quality users.
- The UI must explain why an action is unavailable instead of merely disabling it.

10.3 Security requirements

Area	Requirement
Authentication	OIDC and SAML SSO, MFA through identity provider, session controls and optional local break-glass accounts.
Provisioning	SCIM or API provisioning; immediate deactivation and group mapping.
Authorisation	Server-side enforcement for every request; no client-only security decisions.
Encryption	TLS in transit; strong encryption at rest for databases, object storage and backups.
Tenant isolation	Tenant ID on every record plus database and object-storage controls; automated isolation tests.
Secrets	Managed secret storage, rotation and no secrets in client code or logs.
Secure development	Dependency scanning, code review, static analysis, penetration testing and vulnerability response.
Logging	Security telemetry separated from scientific audit events but correlatable by request ID.
Backups	Encrypted, monitored and regularly restored in a test environment.
Data residency	Region selection where commercially supported; exports and subprocessors documented.

11. Integrations and public API

11.1 Integration principles

- API-first: core product actions are available through the same authorised service layer used by the UI.
- Stable identifiers: integrations use immutable IDs while human-readable IDs remain available.
- Idempotent writes: import and event endpoints support idempotency keys.
- Provenance: imported data retains source system, external ID, time, integration identity and transformation details.
- Least privilege: service accounts use granular scopes and optional project restrictions.
- Observable: imports, webhooks and scheduled jobs expose status, retries, errors and audit links.

11.2 Initial API resources

Resource	Representative operations
Workspaces and projects	List, read, create, update, membership and configuration subject to scope.
Folders and pages	Create, move, read, update properties, export and retrieve versions.
Blocks	Read structured block content, append supported blocks and replace with version checks.
Experiments and protocols	Create, query, update lifecycle, link records and instantiate templates.
Tables and records	Schema, query, bulk upsert, validation, views and export.
Datasets and charts	Create dataset versions, submit analysis jobs, retrieve specifications and exports.
Files	Multipart upload, hash verification, metadata, link and controlled replacement.
Reviews and signatures	Request review, read workflow state and perform authorised transitions.
Audit	Permission-controlled query and export.

11.3 Webhook events

Initial events: project.created, page.updated, experiment.status_changed, review.requested, review.completed, signature.applied, record.locked, file.uploaded, dataset.version_created, import.completed and check.failed. Webhooks are signed, retried with exponential backoff and observable in the administrator interface.

11.4 Target integrations

- Identity providers: Microsoft Entra ID, Okta and generic OIDC/SAML.
- Collaboration: Microsoft Teams, Slack and email.
- Storage and data: S3-compatible storage, SharePoint/OneDrive, data lakehouses and data warehouses.
- Laboratory systems: LIMS, instrument data repositories and vendor export folders through API or controlled agents.
- Developer workflows: GitHub and generic webhooks for protocol/configuration releases where appropriate.
- Reporting: PDF, DOCX, CSV, XLSX and image export; optional BI connectors later.

12. Non-functional requirements

ID	Area	Requirement
NFR-001	Availability	Production service target of 99.9% monthly availability excluding announced maintenance.
NFR-002	Page performance	Typical page shell visible within 2.5 seconds at p95 on a standard corporate connection.
NFR-003	Edit responsiveness	Local editor input response below 100 ms; server acknowledgement normally below 2 seconds.
NFR-004	Table performance	A 10,000-row table view shall filter and sort interactively using server-side processing where required.
NFR-005	Chart performance	Interactive charts shall remain responsive to 100,000 plotted points through aggregation/downsampling.
NFR-006	File scale	Support individual files of at least 5 GB through resumable multipart upload in supported plans.
NFR-007	Concurrency	Support at least 25 concurrent editors on one page without data loss; optimise for 2-10.
NFR-008	Durability	Database and object storage durability shall use managed replication and verified backups.
NFR-009	Recovery	Target RPO <= 15 minutes and RTO <= 4 hours for the standard SaaS service.
NFR-010	Accessibility	Target WCAG 2.2 AA for primary workflows, including keyboard navigation and screen-reader semantics.
NFR-011	Browser support	Current and previous major versions of Chrome, Edge, Firefox and Safari.
NFR-012	Time	Store timestamps in UTC and render in user locale; retain original time-zone context where imported.
NFR-013	Units	Store numeric value and unit separately where unit-aware fields are used.
NFR-014	Audit durability	Audit events shall not be modifiable through application roles or standard APIs.
NFR-015	Observability	Every request and async job shall have a correlation ID; health, latency, error and queue metrics shall be monitored.
NFR-016	Maintainability	Public APIs are versioned; database migrations are reversible or have documented recovery paths.
NFR-017	Portability	Customers can export content, metadata, attachments and structured records in documented formats.
NFR-018	Localisation	Text, date, number and unit formatting shall be localisation-ready.
NFR-019	Privacy	Product telemetry shall avoid scientific content by default and be configurable for regulated deployments.
NFR-020	Print/export	Controlled PDF exports shall be deterministic, paginated and include record identity, version and signatures.

13. Brand and interface system

13.1 Brand positioning

Element	Definition
Name	Sheetbase ELN, shortened to Sheetbase in product UI where context is clear.
Category	Modern electronic lab notebook and scientific data workspace.
Positioning statement	The flexible electronic lab notebook for structured, traceable science.
Primary message	Write naturally. Structure data. Reproduce results.
Tagline option	Structured data. Reliable science.
Personality	Calm, precise, capable, modern and trustworthy.
Avoid	Clinical coldness, oversized marketing typography, playful laboratory cliches, excessive gradients and decorative empty space.

13.2 Visual principles

- Calm density: use compact controls, readable line lengths and visible structure without crowding.

- The document is the hero: navigation and inspectors remain quiet and recede behind the scientific content.
- One primary action colour: blue is reserved for creation, selection and consequential forward actions.
- Status is semantic: green for success/in progress, amber for attention, red for failure, violet for controlled or protocol-related categories.
- Borders before shadows: use 1 px separators and subtle surface changes; shadows are reserved for floating layers.
- Colour is never the only encoding in charts, status or validation messages.

13.3 Colour tokens

Token	Hex	Use
Ink 950	#0F172A	Primary text, logo and high-emphasis icons.
Slate 700	#334155	Secondary text.
Slate 500	#64748B	Metadata and captions.
Slate 300	#CBD5E1	Borders and dividers.
Slate 100	#F1F5F9	Subtle surfaces and selected rows.
White	#FFFFFF	Canvas and primary surfaces.
Blue 600	#2563EB	Primary action and active navigation.
Blue 100	#DBEAFE	Selection, focus and informational surfaces.
Emerald 500	#10B981	Success and active status.
Amber 500	#F59E0B	Warning and due-soon status.
Red 500	#EF4444	Failure, destructive action and critical validation.
Violet 500	#8B5CF6	Protocol, template and controlled-category accent.
Cyan 500	#06B6D4	Secondary data visualisation accent.

13.4 Typography and spacing

Token	Specification	Use
Display	Arial/Inter, 28-32 px, 700	Rare product or project title. Avoid oversized hero text in application screens.
H1	24-28 px, 700	Page title.
H2	18-20 px, 600	Major section title.
H3	14-16 px, 600	Block or card title.
Body	14 px, 400, 1.45 line height	Primary application copy.
Small	12 px, 400	Metadata and helper text.
Data	12-13 px, 400	Tables and dense numeric views.
Monospace	JetBrains Mono or equivalent	Identifiers, code, formulas and raw values.
Grid	8 px base with 4 px half-step	Spacing, component dimensions and layout.

13.5 Component rules

- Buttons: 32 px compact and 36 px standard heights; 6-8 px radius; primary, secondary, tertiary and destructive variants.
- Cards: 1 px border, 8-12 px radius, 16-20 px internal padding; no heavy drop shadow.
- Inputs: persistent label for complex scientific values; units displayed as part of the field, not placeholder text.
- Tables: sticky headers, subtle row hover, selected-row tint and clear focus ring.
- Tags: low-saturation background with high-contrast text; tags should not look like buttons.
- Inspector: 280-340 px width, sectioned cards and consistent property label alignment.
- Icons: simple 1.5 px stroke, rounded joins and familiar metaphors. Avoid decorative laboratory icon overload.
- Dark mode: supported for general use, but controlled print/PDF and comparison exports use the light theme.

13.6 Data visualisation system

- Default categorical sequence: Blue, Emerald, Violet, Amber, Cyan, Rose and Slate.

- Use marker shape and line style in addition to colour for scientific series.
- Default grid lines are light and limited to major intervals.
- Legends use full scientific labels and may include units or batch identifiers.
- Axes must display units and transform state, including log scale.
- Excluded points remain inspectable and visibly distinguished rather than silently removed.
- Every export includes sufficient metadata to identify source chart and revision where configured.

13.7 Voice and terminology

Use	Avoid
Create experiment	Spin up an experiment
Request review	Send it
Changes saved	Magic sync complete
Data integrity check failed	Something went wrong
No results match these filters	Nothing here
This version is locked. Create an amendment to make changes.	You cannot edit this.

14. Reference technical architecture

The following is a recommended architecture, not a mandatory implementation contract. It separates collaborative document editing, relational scientific records, file storage and scientific computation while retaining one coherent permission and audit model.

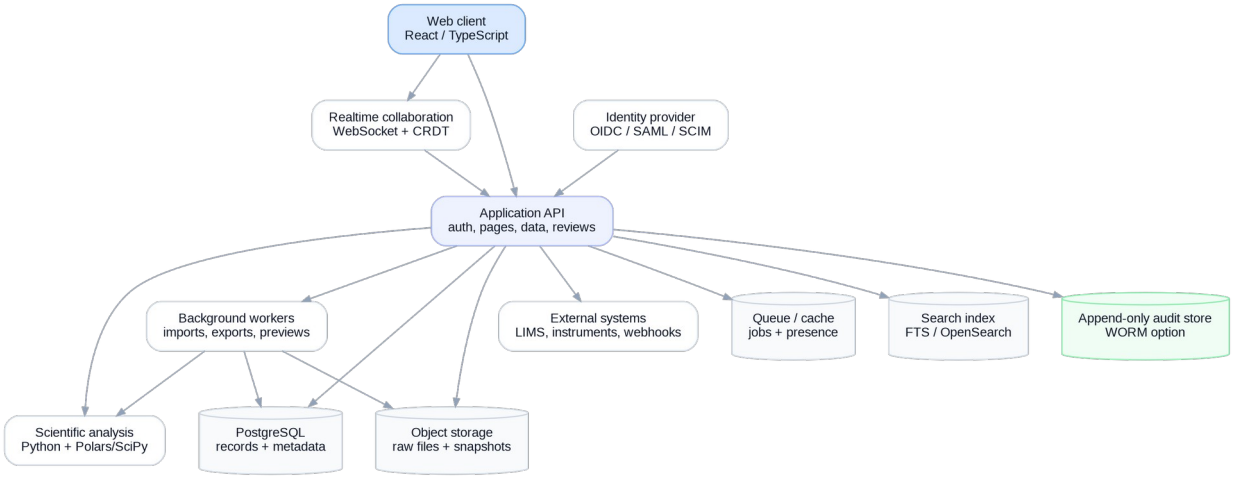


Figure 4. Recommended logical architecture.

14.1 Suggested technology choices

Layer	Recommendation
Web client	React and TypeScript, with Next.js or equivalent routing and rendering framework.
Editor	ProseMirror/Tiptap or equivalent block-capable rich-text engine.
Collaboration	Yjs or equivalent CRDT over WebSockets, persisted as deterministic document updates.
Tables	Virtualised data grid backed by server-side query services.
Charts	Declarative chart specification using Vega-Lite/Plotly/ECharts, with custom scientific extensions.
Application API	Modular service in TypeScript or Python with OpenAPI documentation and strict authorisation middleware.
Scientific analysis	Python workers using Polars/Pandas, NumPy, SciPy and statsmodels as appropriate.
Database	PostgreSQL for records, metadata, permissions, workflows and version indexes.
Object storage	S3-compatible encrypted storage with versioning and lifecycle policies.

Layer	Recommendation
Queue/cache	Redis or managed equivalent for jobs, presence and short-lived cache.
Search	PostgreSQL full-text search initially; OpenSearch when scale or extracted-file search requires it.
Audit	Append-only audit table plus tamper-evident export or WORM archive option.

14.2 Versioning model

- Working state is stored as current relational/document state plus immutable change events.
- Named and controlled versions materialise a deterministic snapshot of page blocks, properties, links, table references and chart specifications.
- Large datasets and files are referenced by immutable version/hash rather than copied into each page version.
- A signature binds to a canonical snapshot hash and its referenced immutable assets.
- Version comparison is produced from structured content, not from rendered PDF differences alone.

14.3 Multi-tenancy and data boundaries

- Every data entity carries a workspace/tenant boundary enforced in database queries and service policy.
- Object-storage keys are tenant-scoped and access is granted through short-lived signed requests.
- Search indexes and background jobs retain tenant context and are covered by automated isolation tests.
- Single-tenant deployment may use dedicated database, storage and encryption keys without changing the application model.

15. Delivery plan and success measures

15.1 Recommended release sequence

Release	Scope	Exit outcome
Foundation prototype	Application shell, project tree, notebook editor, inline tables, file blocks and design system.	Validate core interaction and information density.
Controlled beta	Experiments, shared tables, version history, comments, permissions, audit events, review and basic signatures.	Prove end-to-end controlled record workflow.
Production R1	Analysis Studio, chart blocks, protocols/templates, search, exports, API, admin, performance and validation support.	Launch a credible ELN for research and controlled pilots.
R2 structured science	Sample registry, unit-aware fields, curve fitting, dashboards, automations, advanced retention and integration connectors.	Increase reuse and operational visibility.
Future platform	Custom record types, configurable navigation, plug-in analysis, mobile scanning, instrument agents and carefully governed AI.	Expand from ELN to extensible scientific workspace.

15.2 Product success measures

Measure	Definition
Activation	Percentage of invited users who create or materially edit an experiment within seven days.
Adoption	Weekly active scientists and active projects; percentage of experiments recorded in Sheetbase.
Completion	Experiments reaching completed/approved state with required evidence.
Efficiency	Median time from experiment completion to review submission and approval.
Data reuse	Percentage of charts and datasets reused across more than one page or project.
Quality	Rate of failed integrity checks, missing required links and review returns.
Reliability	Availability, save latency, sync conflict rate and failed background jobs.
Portability	Successful export-package generation and restore/import testing.
User sentiment	Task-level satisfaction for recording, finding, analysing and reviewing work.

15.3 Initial research and usability validation

11. Observe scientists creating a real experiment from a blank page and from a template.

12. Test whether users understand inline versus shared tables and can promote data confidently.
13. Test chart creation from a table, including log axes, error bars and exclusion reasons.
14. Test project/folder navigation with at least three concurrent projects and 50+ items.
15. Run a complete review, version comparison, signature and amendment scenario with QA users.
16. Measure the ability to reconstruct source file, calculation and chart lineage without facilitator help.
17. Test a controlled export package with an independent reviewer.

16. Risks, assumptions and open decisions

16.1 Key risks and mitigations

Risk	Impact	Mitigation
Scope expansion into LIMS and statistics suite	The product could become too broad to execute.	Keep the page, tables, charts and controlled workflow as the core; use integrations and phased delivery.
Collaborative editing versus auditability	CRDT operations may be difficult to present as meaningful changes.	Audit user-level actions and controlled snapshots; keep raw collaboration operations as technical evidence.
Scientific calculation correctness	Incorrect formulas or fits could undermine trust.	Use tested libraries, version calculation engines, publish methods and include verification datasets.
Over-customisation	Unbounded schemas and workflows can harm usability and validation.	Curate core record types and expose configuration progressively with safe defaults.
Large data in a document UI	Huge files and tables may degrade performance.	Reference immutable datasets, virtualise views and use background/server-side computation.
Regulatory overclaim	Marketing may imply compliance or validation.	Use precise language: supports controlled use and validation; provide evidence, not certification claims.
Vendor lock-in concerns	Customers may resist storing critical science in a proprietary format.	Document schemas, provide complete exports and maintain a stable API.
Weak information density	Generic SaaS styling may waste space and frustrate scientists.	Apply calm-density design rules and test on real notebook content.

16.2 Assumptions

- Most users work primarily on desktop or laptop computers with a keyboard and mouse.
- Experiments benefit from structured metadata but do not require a large universal mandatory schema.
- Scientific organisations will continue to use specialist systems; Sheetbase must link and integrate rather than replace all of them.
- Customers requiring regulated use will perform intended-use assessment, validation and procedural control.
- The first release can depend on network connectivity for controlled operations.

16.3 Open product decisions

Decision	Recommended default	Decision point
Product name	Use Sheetbase ELN externally and Sheetbase in the app.	Before public brand launch.
Deployment options	SaaS first, single-tenant private cloud as enterprise option.	Before regulated design partners commit.
Advanced statistics	Routine scientific statistics in R1/R2; specialist analyses through API/plugins.	During Analysis Studio technical discovery.
Offline mode	Local resilience only in R1; no controlled offline signing.	After field and lab connectivity research.
Custom object builder	Defer until core objects and workflows are stable.	After R1 adoption data.
Mobile app	Responsive viewing first; scanning/capture application later.	After sample-registry validation.
AI features	No autonomous scientific conclusions; permission-aware assistive features only.	Separate risk and governance review.

17. Product acceptance criteria

The R1 product baseline is acceptable when the following end-to-end scenarios pass in a production-like environment with representative scientific data and permissions.

18. A workspace administrator can create teams, a project and a nested project folder structure, then assign role-based access.
19. A scientist can create an experiment with a stable ID and free-form notebook page from a template.
20. The scientist can add text, sections, images, files, an inline table and a chart without leaving the page.
21. An inline table can be promoted to a shared table and the page continues to show the correct linked view.
22. A chart can be created from the table with log scale, series grouping and error bars, then exported as SVG and PNG.
23. The chart retains its source version, transformations, settings and analysis engine version.
24. The experiment can link a sample, protocol, dataset and raw instrument file with visible provenance.
25. Two authorised users can edit and comment concurrently without losing content or authorship.
26. The author can submit a stable version for review and the reviewer can compare it with the previous version.
27. The reviewer can request changes; the author can revise and resubmit without changing the original review target.
28. Required reviewers can electronically sign the approved version and the system can lock it.
29. An authorised amendment can create a new version while the prior signed version remains intact and retrievable.
30. The audit explorer can reconstruct creation, edits, file upload, submission, comments, signatures, lock and export.
31. A complete export package contains a readable record, structured data, attachments, signatures, audit trail and hash manifest.
32. A viewer without edit or export permission cannot bypass restrictions through the UI or API.
33. Search returns authorised matches across pages, identifiers and table values without leaking inaccessible content.
34. Backup restoration and tenant-isolation tests pass according to the non-functional requirements.

Appendix A. Default statuses and meanings

Status	Meaning	Editing
Draft	Record created but work has not formally begun.	Authorised editors.
In progress	Active scientific work.	Authorised editors.
Awaiting review	Stable version under review.	Review target immutable; new draft allowed.
Changes requested	Review returned to author.	Authorised editors on new draft.
Approved	Required review and signatures complete.	No content edit.
Locked	Final controlled record.	Amendment only.
Superseded	Replaced by a newer controlled version.	Read-only.
Archived	Retained but removed from active navigation.	Read-only unless restored by policy.

Appendix B. Illustrative permission matrix

Capability	Admin	QA	Lead	Scientist	Reviewer	Viewer
View authorised content	Yes	Yes	Yes	Yes	Assigned	Yes
Create/edit experiment	Config	Yes	Yes	Yes	No	No
Manage project folders	Yes	Yes	Yes	Limited	No	No
Submit for review	Yes	Yes	Yes	Yes	No	No
Review/comment	Yes	Yes	Yes	Optional	Yes	No
Approve/sign	Policy	Yes	Policy	Policy	Assigned	No
Lock/unlock	Policy	Yes	Policy	No	No	No
Manage templates	Yes	Yes	Yes	Optional	No	No
Export controlled package	Policy	Yes	Yes	Policy	Policy	Policy
Manage users/integrations	Yes	Limited	No	No	No	No

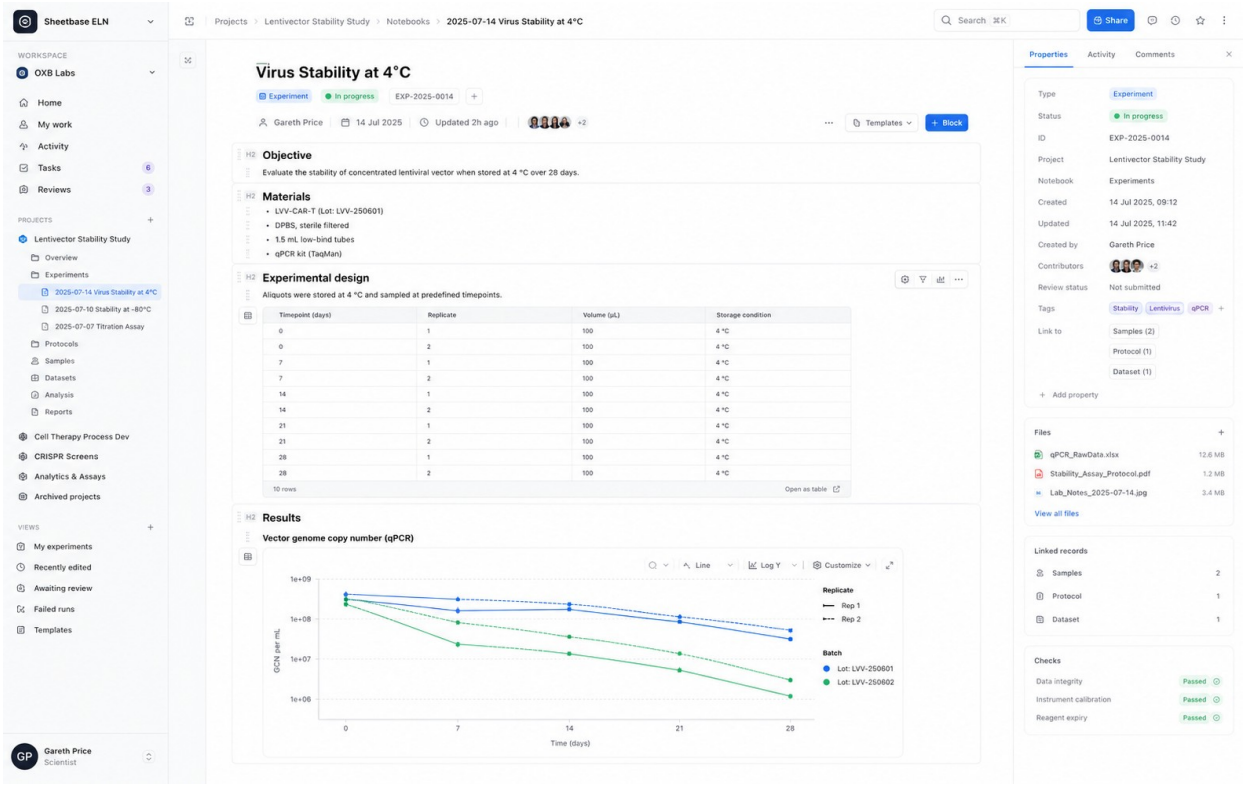
Capability	Admin	QA	Lead	Scientist	Reviewer	Viewer
View audit trail	Yes	Yes	Project	Own scope	Review scope	No

Appendix C. Core export package

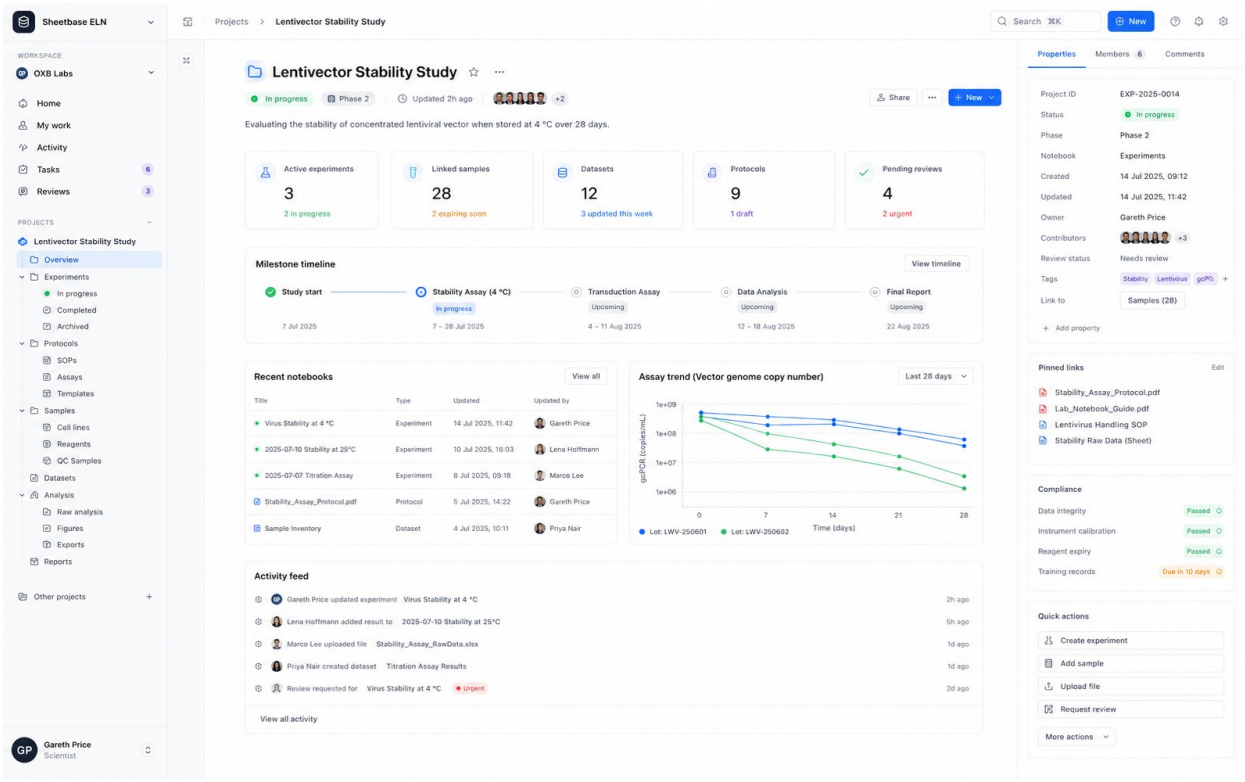
- record.pdf - deterministic human-readable record including metadata, version, signatures and linked-item summary.
- record.json - structured page, block and property representation.
- tables/ - CSV or Parquet files for included tables and datasets, with schema metadata.
- charts/ - chart specifications and rendered exports.
- files/ - original attachments using safe names plus original-name metadata.
- audit.csv and audit.json - authorised audit trail extract.
- manifest.json - entity IDs, versions, relationships, checksums and export metadata.
- README.txt - package structure and verification instructions.

Appendix D. Visual concept references

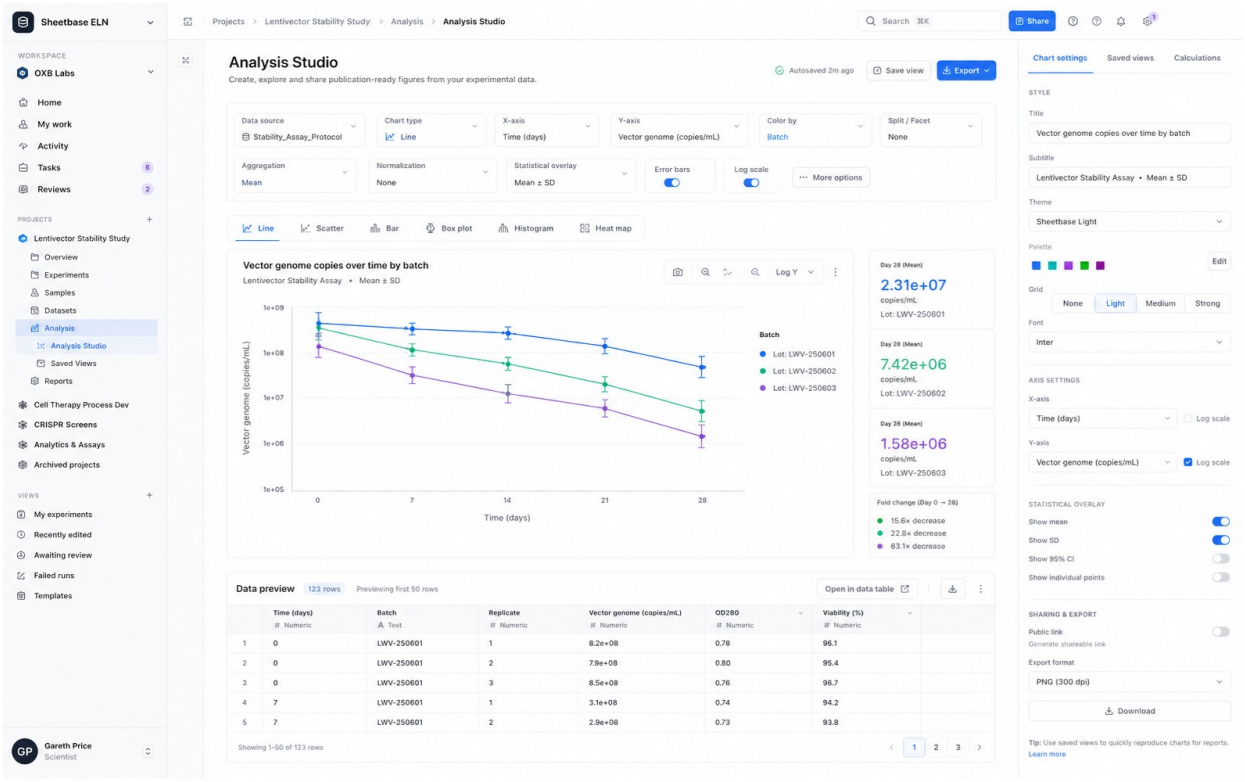
The following visuals are directional concepts only. The product specification and design system in this document are authoritative where the imagery contains inconsistencies.



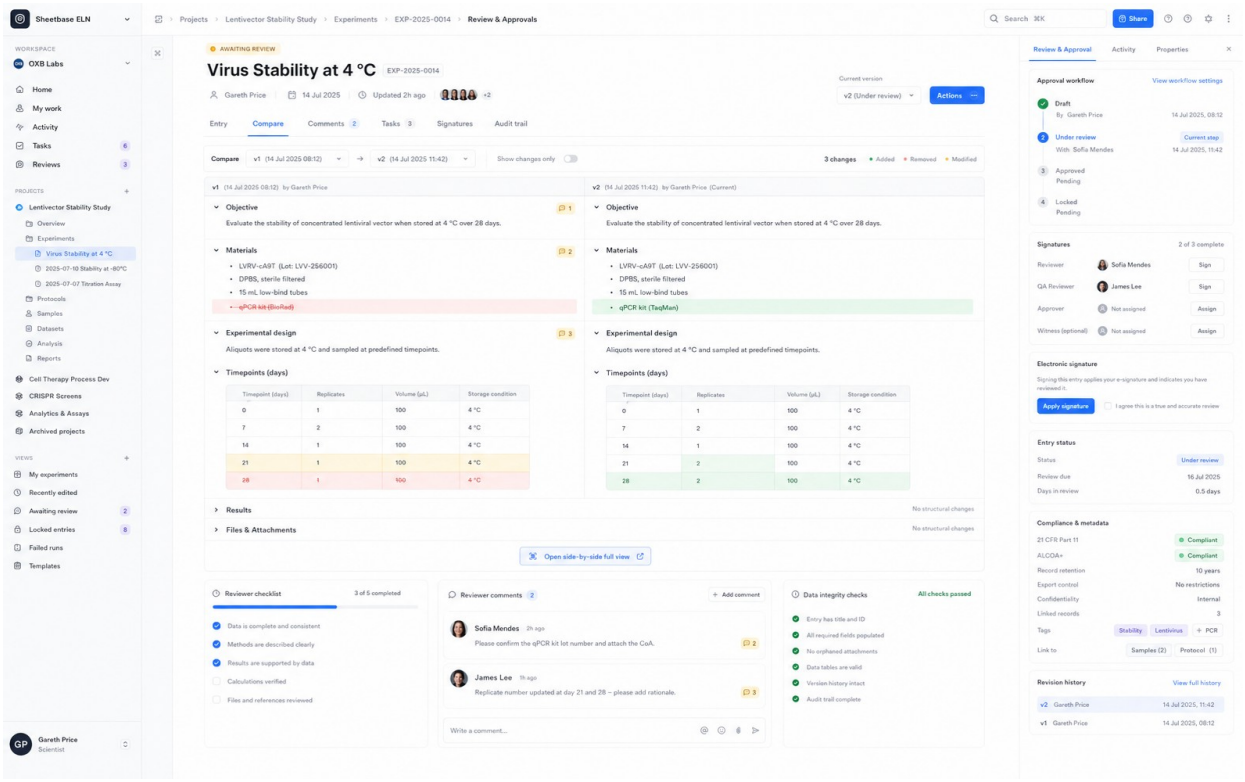
Notebook page concept



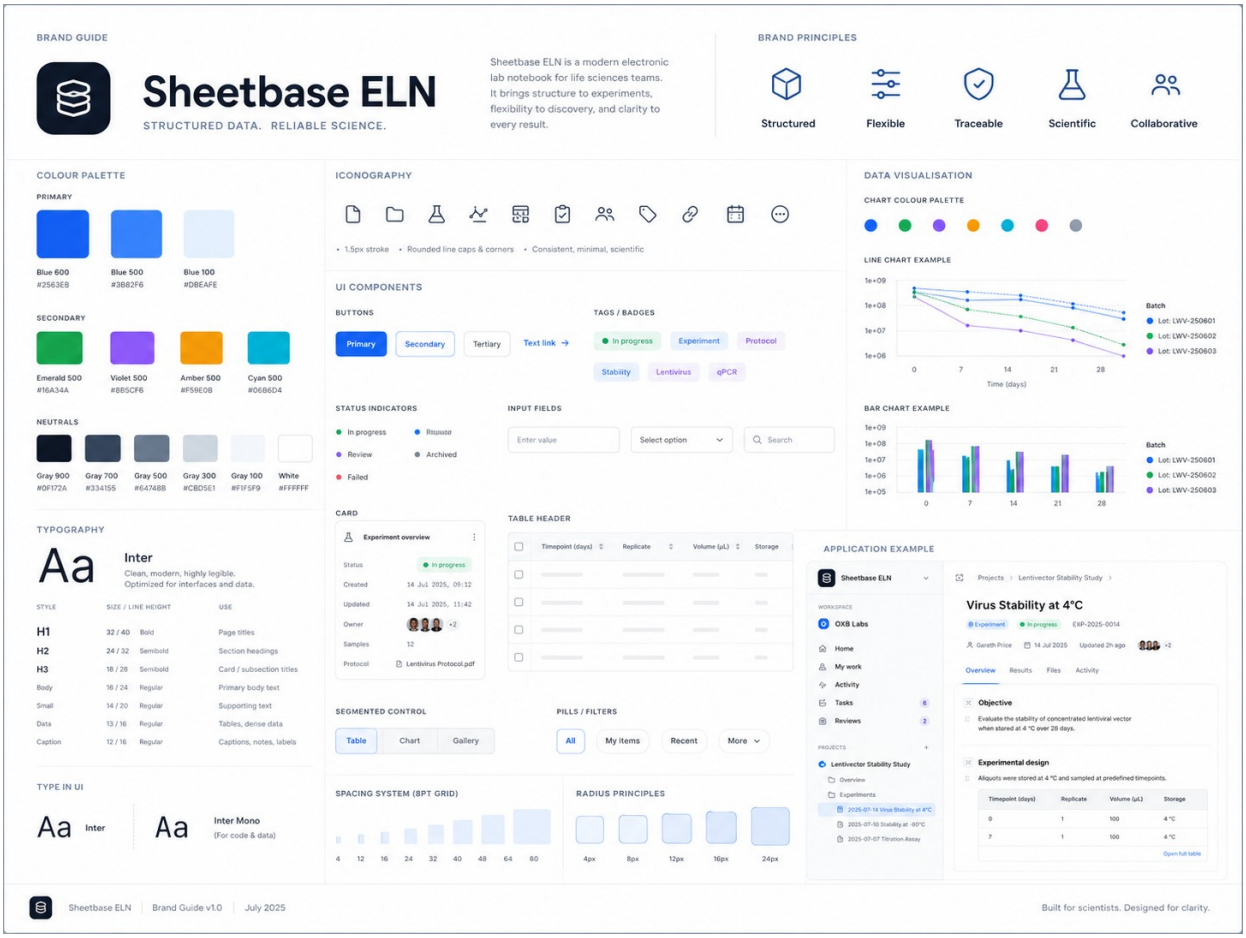
Project overview concept



Analysis Studio concept



Review and version comparison concept



Initial brand system concept

Appendix E. Regulatory and guidance references

References are included to explain the design baseline and should be reviewed by qualified quality and legal personnel for each intended use and jurisdiction.

R1. Electronic Code of Federal Regulations, Title 21, Part 11 - Electronic Records; Electronic Signatures. Official source

R2. US FDA, Part 11, Electronic Records; Electronic Signatures - Scope and Application. Official source

R3. European Commission, EudraLex Volume 4, Annex 11 - Computerised Systems (current published baseline listed as revision January 2011). Official source

R4. European Commission, 2025 stakeholder consultation on revised Chapter 4 and Annex 11. Official source

R5. MHRA, GxP Data Integrity Guidance and Definitions. Official source